

Reply to the Referee Report

JHEP_282P_0726: *Soft Gluon Wave Function and Evolution Operator in the CGC at Next-to-Leading Order*

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How this document is organised. Part I answers the referee's four comments one by one. Each answer quotes the comment, gives the reply, and lists the changes to make in the manuscript. Part II lists further errors found in a line-by-line check of the manuscript. Each entry names the equation, what is printed, the correction, and how it was checked. Equation numbers refer to the submitted version (v1). Orange boxes are notes to the author and should be removed before the reply is sent.

Note to the author (not for the referee)

Two kinds of statement appear below. *Derived*: re-derived here from the paper's own matrix elements (App. D) and state normalisations (App. B), or checked by dimensions. *To verify*: plausible but depends on conventions that could not be fully reconstructed from the PDF. Every *To verify* item should be checked against your own notes before you submit. The new t -channel expression in Part I is written in terms of the paper's matrix elements. Its final (2π) factors and colour/argument labels must be fixed in the conventions of Eq. (5.7).

Dear Editor,

I thank the referee for a careful reading of the manuscript and for comments that improve the paper substantially. Below I reply to each comment and describe the changes made. In the course of the revision I also re-checked the whole manuscript and corrected a number of further errors, listed in Part II.

Part I — Reply to the referee

Comment 1: the t -channel gluon exchange in Sec. 4.3.1

Referee, comment 1

It seems that in Sec 4.3.1. one is neglecting a “ t -channel gluon exchange” contribution, where one passes through a three-gluon intermediate state. This is a significant part of the result. It is also curious that the authors do not notice the absence of this contribution in the unitarity relations, which should not hold when it is neglected.

Reply. The referee is correct. Eq. (4.82) contains all three channels, but only for the interactions that are a single $\mathcal{O}(g^2)$ vertex. Besides the four-gluon contact term, its three colour structures carry the *instantaneous* gluon exchange in the three channels:

- $f^{a'ad} f^{a'cb}$: t -channel ($k \rightarrow r$), denominator $(k^+ - r^+)(p^+ - q^+) = -(k^+ - r^+)^2$;
- $f^{a'ac} f^{a'db}$: u -channel ($k \rightarrow q$), denominator $(k^+ - q^+)(p^+ - r^+) = -(k^+ - q^+)^2$;
- $f^{a'ab} f^{a'dc}$: s -channel, denominator $(k^+ + p^+)(r^+ + q^+) = (k^+ + p^+)^2$.

In light-cone perturbation theory the exchange of a gluon has two parts. The instantaneous part is a single vertex of $H_{ggg\text{-inst}}$. The propagating part is second order in H_{ggg} and passes through an intermediate state. For the s -channel the paper includes both: the instantaneous part in Eq. (4.82) and the propagating part, with a one-gluon intermediate state, in Eq. (4.84). For the t - and

u -channels only the instantaneous part was included. The propagating part was missed: one incoming gluon emits a gluon through H_{ggg} and the other incoming gluon absorbs it through a second H_{ggg} , so the intermediate state has three gluons. This is the contribution the referee refers to. It is evaluated below and added to the paper.

The missing diagram. Label the incoming gluons $g_i^a(k)$, $g_j^b(p)$ and the outgoing gluons $g_l^d(r)$, $g_k^c(q)$, with $k+p=r+q$. The exchanged gluon r' must have positive plus momentum. Two light-cone time orderings are therefore possible, and for given external momenta exactly one of them contributes:

- $k^+ > r^+$: $k \rightarrow r + r'$ first, then $r' + p \rightarrow q$. The intermediate state is $|g(r)g(r')g(p)\rangle$ with $r' = k - r$.
- $k^+ < r^+$: $p \rightarrow q + r''$ first, then $r'' + k \rightarrow r$. The intermediate state is $|g(k)g(r'')g(q)\rangle$ with $r'' = p - q$.

The two orderings are shown in Fig. 1(a),(b), together with the s -channel and instantaneous exchanges already in the paper, (c),(d).

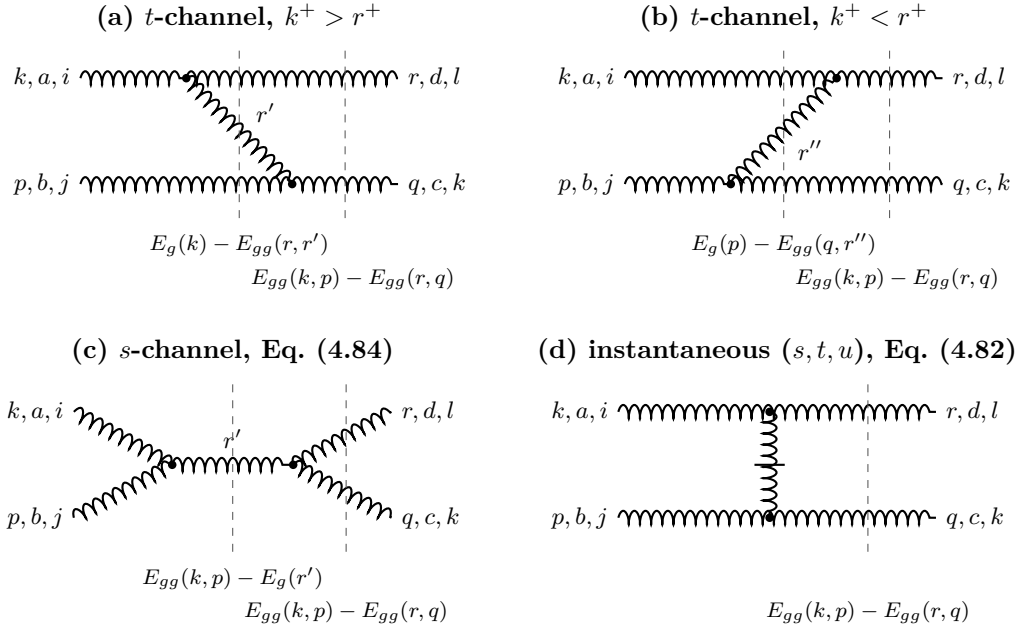


Figure 1: Gluon exchange between two incoming soft gluons at $\mathcal{O}(g^2)$. Light-cone time runs from left to right; dashed vertical lines mark the intermediate and final states, with the corresponding energy denominators written below. Diagrams (a) and (b) are the two light-cone time orderings of the non-instantaneous t -channel exchange, which are missing from Sec. 4.3.1; their intermediate state has *three* gluons. Plus-momentum positivity of the exchanged gluon ($r'^+ = k^+ - r^+ > 0$ in (a), $r''^+ = r^+ - k^+ > 0$ in (b)) means that exactly one of them contributes for given external momenta. The u -channel diagrams follow from $r \leftrightarrow q$. For comparison, (c) is the s -channel exchange already contained in Eq. (4.84), and (d) is the instantaneous gluon exchange (crossed line). Eq. (4.82) contains the instantaneous exchange in all three channels, one per colour structure, but not the propagating t - and u -channel exchanges (a), (b).

The u -channel contribution is obtained from the exchange $r \leftrightarrow q$ (with $d \leftrightarrow c$, $l \leftrightarrow k$). In LCPT,

$$\begin{aligned} |(\Psi_{ji}^{ba}(k, p))_{gg}^t\rangle &= \frac{1}{2} \int d^3r d^3q |g_l^d(r) g_k^c(q)\rangle \frac{1}{E_{gg}(k, p) - E_{gg}(r, q)} \\ &\times \left[\int d^3r' \frac{\langle g_l^d(r) g_k^c(q) | H_{ggg} | g_l^d(r) g_m^e(r') g_j^b(p) \rangle}{E_g(k) - E_{gg}(r, r')} \right. \\ &\quad \left. \times \langle g_l^d(r) g_m^e(r') g_j^b(p) | H_{ggg} | g_i^a(k) g_j^b(p) \rangle \right]_{\text{conn}} + (k \leftrightarrow p), \end{aligned} \quad (1)$$

Here “conn” keeps only the contractions in which the gluon emitted by one incoming line is absorbed by the other. Both vertices are given by Eq. (D.3). The spectator’s energy cancels in the intermediate denominator:

$$E_{gg}(k, p) - E_{ggg}(r, r', p) = E_g(k) - E_{gg}(r, r').$$

This denominator vanishes only at the collinear endpoint of the $k \rightarrow r r'$ splitting, so it is the same (integrable) denominator that appears in $\mathbb{C}$, Eq. (5.5).

Expression in terms of the Ω coefficients. Write $\omega_1(f \leftarrow i) = V_{fi}/(E_i - E_f)$ for the first-order LCPT amplitude and use $V_{fm} = \omega_1(f \leftarrow m)(E_m - E_f)$. Every second-order amplitude with a single intermediate state m can then be written as

$$\omega_2(f \leftarrow i) = \sum_m \omega_1(f \leftarrow m) \omega_1(m \leftarrow i) \frac{E_m - E_f}{E_i - E_f}. \quad (2)$$

Here $\omega_1(rr' \leftarrow k)$ is the splitting coefficient $\mathbb{C}$ of Eq. (5.5), and $\omega_1(q \leftarrow r'p)$ is the merging coefficient $\mathbb{C}_2 = -\mathbb{C}^\dagger$ of Eq. (4.8). The new contribution to $\mathbb{D}_2$ is therefore

$$\mathbb{D}_2^{(t)} \propto \int \frac{d^3r'}{(2\pi)^3} \omega_1(q \leftarrow r'p) \omega_1(rr' \leftarrow k) \frac{E_g(r') + E_g(p) - E_g(q)}{E_{gg}(k, p) - E_{gg}(r, q)} + (k \leftrightarrow p) + (r \leftrightarrow q), \quad (3)$$

with the same (2π) normalisation as the s -channel $\mathbb{C}\mathbb{C}^\dagger$ term already in Eq. (5.7). This term is ρ -independent, so the ordering of $\mathbb{C}$ and $\mathbb{C}^\dagger$ does not matter here.

Why unitarity signals the omission. The referee is right that the unitarity relation Eq. (4.11) cannot hold without this term. From Eq. (2) and its Hermitian conjugate with $i \leftrightarrow f$, for each intermediate state m one finds

$$\begin{aligned} \omega_2(f \leftarrow i) + [\omega_2(i \leftarrow f)]^\dagger &= \frac{V_{fm}V_{mi}}{E_i - E_f} \left[\frac{1}{E_i - E_m} - \frac{1}{E_f - E_m} \right] = -\frac{V_{fm}V_{mi}}{(E_i - E_m)(E_f - E_m)} \\ &= -[\omega_1(m \leftarrow f)]^\dagger \omega_1(m \leftarrow i). \end{aligned} \quad (4)$$

This is the $\mathcal{O}(g^2)$ unitarity condition Eq. (4.5), $\Omega^{(2)} + \Omega^{(2)\dagger} = -\Omega^{(1)\dagger}\Omega^{(1)}$, and it holds *intermediate state by intermediate state*. The ordering $V_{fm}V_{mi}$ is kept throughout, so the identity also holds for the non-commuting, ρ -dependent coefficients.

The right-hand side of Eq. (4.11) contains $\mathbb{C}^\dagger\mathbb{C}$ products in which the internal gluon connects *different* external gluons, i.e. intermediate states of the t -channel type $|g(r)g(r')g(p)\rangle$. By Eq. (4) these terms can only be reproduced by the Hermitian part of a second-order diagram with the same intermediate state, which is exactly the t -channel diagram. The $\mathbb{D}_2$ of Eq. (5.7) contains only the s -channel $\mathbb{C}\mathbb{C}^\dagger$ product and the $\mathbb{A}\mathbb{C}$ products, so it cannot satisfy Eq. (4.11).

In the submitted version Eq. (4.11) was not explicitly checked against Eq. (5.7); this was my oversight. In the revised version the check is done term by term using Eq. (4). For every intermediate state (vacuum, one-, two- and three-gluon) it shows that the matched coefficients satisfy (4) once the t - and u -channel terms are included.

Consequences elsewhere. The same exchange appears as a disconnected piece (third gluon as spectator) in the three-gluon sector Eq. (4.18). It is accounted for automatically because the normal-ordered operator $\mathbb{D}_2 a^\dagger a^\dagger a a$ acts on every pair. In Sec. 6, the new term in $\mathbb{D}_2$ is needed to cancel the part of $\frac{1}{2}[H_{ggg}, \Omega^{(1)}]$ with t -channel structure in the $2 \rightarrow 2$ sector (see also Part II, item A3).

Proposed changes to the manuscript

1. Sec. 4.3.1: after Eq. (4.82), state that its three colour structures contain the *instantaneous* exchange in the t -, u - and s -channels. Add the propagating t - and u -channel exchange diagrams (with figure), Eq. (1), and their evaluated form. Change “six diagrams” accordingly.
2. Eq. (4.95): add $|\langle \Psi_{ji}^{ba}(k, p) \rangle_{gg}^t \rangle$. Eq. (5.7): add $\mathbb{D}_2^{(t)}$ of Eq. (3).
3. Sec. 4 (after Eq. (4.14)) and Sec. 5: state that $\mathbb{B}_3$ and $\mathbb{D}_2$ are fully fixed by the matching. The unitarity relations (4.10), (4.11) are then *checks*, not determinations. Add the general identity Eq. (4) and state explicitly that the checks are satisfied.
4. Correct the sentence after Eq. (5.12) and the list on p. 14 (“The coefficients $\mathbb{A}_1, \dots$ are determined by matching”) so that they agree with item 3.

Note to the author (not for the referee)

On whether (4.82) already contains this: it contains all three channels, but only their instantaneous parts. (4.82) is a single vertex, so it has no intermediate state, whereas the referee explicitly describes a contribution that “passes through a three-gluon intermediate state”. Your own (4.84) is the s -channel version of exactly this kind of term (two H_{ggg} vertices, one-gluon intermediate state). The t - and u -channel versions, with three-gluon intermediate states, do not appear among the six diagrams of Sec. 4.3.1. Separately, Eq. (4) shows that (4.11) cannot be satisfied without them. So the referee’s point should be conceded, as the reply does.

Comment 2: vanishing $2g \rightarrow 2g$ energy denominator

Referee, comment 2

This is related to both the “ t -channel gluon exchange” contribution mentioned above and the contribution already included in (4.82). The $2g \rightarrow 2g$ energy denominator can go to zero and thus needs to be regularized. How this is done should be discussed in the manuscript.

Reply. I thank the referee. The overall energy denominator can indeed vanish, and the revised manuscript states explicitly how it is treated.

Why the denominator can vanish. The δ -function in Eq. (4.82) conserves the plus and transverse momenta, but not the light-cone energy; non-conservation of p^- at the vertices is what produces energy denominators in LCPT. After using it, $q = k + p - r$, the denominator is still a function of the free outgoing momentum r :

$$D(r) = \frac{\mathbf{k}^2}{2k^+} + \frac{\mathbf{p}^2}{2p^+} - \frac{\mathbf{r}^2}{2r^+} - \frac{(\mathbf{k} + \mathbf{p} - \mathbf{r})^2}{2(k^+ + p^+ - r^+)}.$$

This vanishes at $r = p$ (so $q = k$), which lies inside the integration region, while the numerator of Eq. (4.82) stays finite there. More generally, writing $E_{gg}(k, p) = \mathbf{P}^2/2P^+ + \boldsymbol{\kappa}^2/[2P^+\xi(1-\xi)]$ with $\xi = k^+/P^+$ and $\boldsymbol{\kappa} = (1-\xi)\mathbf{k} - \xi\mathbf{p}$, the denominator vanishes on the whole surface

$$\frac{\boldsymbol{\kappa}^2}{\xi(1-\xi)} = \frac{\boldsymbol{\kappa}'^2}{\xi'(1-\xi')},$$

see Fig. 2(a),(c). This cannot happen for the vacuum incoming state of Sec. 4.1, where $E_i = 0 < E_f$, which is why the issue first appears with incoming soft gluons.

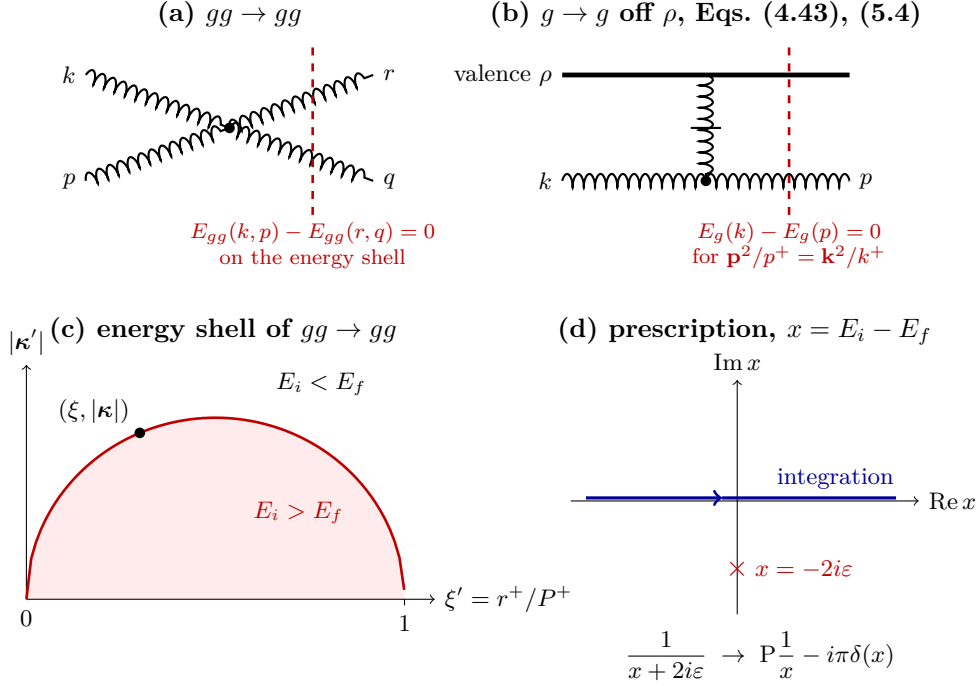


Figure 2: The vanishing energy denominator of Comment 2. (a) Any $gg \rightarrow gg$ amplitude, whether contact, instantaneous, s - or t -channel, carries the overall denominator $E_{gg}(k, p) - E_{gg}(r, q)$ of the final state (red dashed cut). The δ -function fixes $q = k + p - r$ but not the energy, and the denominator vanishes e.g. at $r = p$, $q = k$. (b) The same happens for gluon-number-preserving scattering off the valence background, e.g. the instantaneous term (4.43) and $\mathbb{B}_3$ in (5.4). (c) With $\xi = k^+/P^+$ and $\kappa = (1 - \xi)\mathbf{k} - \xi\mathbf{p}$ fixed, the denominator vanishes on the curve $|\kappa'|^2 = |\kappa|^2 \xi'(1 - \xi')/[\xi(1 - \xi)]$ (red), which lies *inside* the region of integration over the outgoing momenta; it separates configurations with $E_i > E_f$ (shaded) from those with $E_i < E_f$. (d) The adiabatic switching of Eq. (3.7) moves the pole to $x = -2i\varepsilon$ below the real axis at second order. As $\varepsilon \rightarrow 0$ this gives a principal value, which is integrable, plus an on-shell δ -function term.

Which coefficients are affected. Splitting a gluon or emitting one from the valence raises the free light-cone energy; merging or absorbing lowers it. The overall denominator $E_i - E_f$ can therefore vanish only for processes that combine both, or for elastic scattering:

Process	Equations	Coefficient	Can $E_i - E_f$ vanish?
$1 \rightarrow 1$ off the background	(4.43)–(4.55)	$\mathbb{B}_3$	yes
$2 \rightarrow 2$	(4.82), (4.84), (4.86)–(4.92), t/u -channel	$\mathbb{D}_2$	yes
$3 \rightarrow 3$	(4.97), (4.99)	$\mathbb{E}_2$	yes
vacuum sector	Sec. 4.1	$\mathbb{N}, \mathbb{A}, \mathbb{B}_2$	no ($E_i = 0$)
splitting, emission + splitting	(4.57), (4.59), (4.61), (4.63), (4.65)	$\mathbb{C}, \mathbb{D}_1$	no (collinear endpoints only)
$2 \rightarrow 4$	(4.94)	$\mathbb{E}_1$	no (pole cancels, see below)

In Eq. (4.94) the overall pole cancels between the two time orderings: with $A = E_g(k) - E_{gg}(r, s) \leq 0$ and $B = E_g(p) - E_{gg}(u, v) \leq 0$ one has $\frac{1}{A+B}(\frac{1}{A} + \frac{1}{B}) = \frac{1}{AB}$, in agreement with $\mathbb{E}_1 \propto \mathbb{C}\mathbb{C}$, Eq. (5.8).

Intermediate-state denominators, e.g. $E_g(k) - E_{gg}(r, r')$ in the t -channel or $E_{gg}(k, p) - E_g(r')$ in Eq. (4.84), involve a single splitting or merging and vanish at most at collinear endpoints, where they are integrable. The coefficients fixed by unitarity, $\mathbb{B}_1$, $\mathbb{D}_3$ and $\mathbb{E}_3$ (Eqs. (4.9), (4.12), (4.14)), are built from $\mathbb{A}$, $\mathbb{B}_2$, $\mathbb{C}$, $\mathbb{D}_1$, $\mathbb{E}_1$ and are unaffected. Only $\mathbb{B}_3$, $\mathbb{D}_2$ and $\mathbb{E}_2$ need a prescription.

The prescription. The following paragraph is added to Sec. 3.2:

Proposed changes to the manuscript

“We use non-degenerate (Rayleigh–Schrödinger) perturbation theory. For the continuum of free soft-gluon states, the degenerate configurations $E_i = E_f$ form a set of measure zero, and the perturbative series is defined in the standard way by the prescription $E_i - E_f \rightarrow E_i - E_f + i\varepsilon$ at n -th order. This prescription follows from the adiabatic definition of Ω , Eq. (3.7). With it the unitarity relations (4.4)–(4.5) hold identically. The denominators that can vanish occur only in $\mathbb{B}_3$, $\mathbb{D}_2$ and $\mathbb{E}_2$; the $\mathcal{O}(g)$ coefficients $\mathbb{A}$ and $\mathbb{C}$ are unaffected. Couplings between exactly degenerate states cannot be removed by a unitary transformation, so the transformed Hamiltonian $\Omega^\dagger H \Omega$ commutes with H_0 .”

Explicitly, with the interaction switched on adiabatically, $H_I(x^+) e^{\varepsilon x^+}$, the first- and second-order amplitudes are

$$\omega_1(f \leftarrow i) = \frac{V_{fi}}{E_i - E_f + i\varepsilon}, \quad \omega_2(f \leftarrow i) = \sum_m \frac{V_{fm} V_{mi}}{(E_i - E_f + 2i\varepsilon)(E_i - E_m + i\varepsilon)}, \quad (5)$$

and a short calculation, analogous to Eq. (4), gives for every $\varepsilon > 0$

$$\begin{aligned} \omega_1(i \leftarrow f)^\dagger &= -\omega_1(f \leftarrow i), \\ \omega_2(f \leftarrow i) + \omega_2(i \leftarrow f)^\dagger &= -\sum_m \frac{V_{fm} V_{mi}}{(E_i - E_m + i\varepsilon)(E_f - E_m - i\varepsilon)} = -\sum_m \omega_1(m \leftarrow f)^\dagger \omega_1(m \leftarrow i). \end{aligned}$$

These are the unitarity conditions (4.4), (4.5); the $2i\varepsilon$ at second order is what makes the second identity exact. As $\varepsilon \rightarrow 0$, $1/(x + 2i\varepsilon) \rightarrow \mathcal{P}(1/x) - i\pi\delta(x)$ with $x = E_i - E_f$ (Fig. 2(d)). The principal value is integrable, since the pole is simple. The δ -function term is an on-shell, anti-Hermitian contribution, which does not enter the unitarity relations. The adiabatic construction of unitary light-cone perturbation theory was discussed recently in Ref. [41], which I now cite here.

Consequence for Sec. 6. At $\mathcal{O}(g)$ nothing changes: $\mathbb{A}$ and $\mathbb{C}$ have no vanishing denominators, and Eq. (6.5) holds as stated. At $\mathcal{O}(g^2)$, Ω removes all couplings between states of different free light-cone energy. The energy-conserving $2 \rightarrow 2$ and background $1 \rightarrow 1$ couplings remain, so the statement “ $H_{\text{diag}} = \sum_n E_n |n\rangle \langle n|$ ” is replaced by $[\Omega^\dagger H \Omega, H_0] = 0$, i.e. $\Omega^\dagger H \Omega$ is block-diagonal in the free light-cone energy. No coefficient computed in the paper changes, apart from the specification of the prescription.

Proposed changes to the manuscript

1. Sec. 3.2, after Eq. (3.5): add the paragraph quoted above.
2. Sec. 3.3: write Eq. (3.7) in the interaction picture, with light-cone time x^+ and the switching factor $e^{\varepsilon x^+}$, and give Eq. (5) with the unitarity identities.
3. After Eqs. (4.43) and (4.82): one sentence noting that the overall denominator vanishes for degenerate configurations (e.g. $r = p$, $q = k$ in (4.82)) and is understood with the prescription of Sec. 3.2.
4. Sec. 5: state that $\mathbb{B}_3$, $\mathbb{D}_2$, $\mathbb{E}_2$ carry this prescription and that all other coefficients are free of vanishing denominators.
5. Sec. 6, abstract and conclusions: replace “diagonal” at $\mathcal{O}(g^2)$ by “commutes with H_0 ”

(block-diagonal in free light-cone energy), and mention the on-shell remainder.

Note to the author (not for the referee)

Why not answer that the paper “uses non-degenerate perturbation theory, so the comment does not apply”: non-degenerate perturbation theory assumes that no denominator vanishes, and the free two-gluon spectrum is a continuum in which the forward point $r = p$, $q = k$ is always degenerate with the initial state. Choosing the method does not remove the degeneracy, and the referee would read this as conceding the point. The paragraph above keeps the non-degenerate framing, which is standard for continuum states, and adds the one ingredient the referee asks for: the prescription. Without it, the coefficients $\mathbb{B}_3$, $\mathbb{D}_2$, $\mathbb{E}_2$ are not well-defined functions, and the norm of the state, which involves $|D|^{-2}$, is not integrable across $D = 0$. If you prefer the principal value instead of $i\varepsilon$, it also satisfies the unitarity identities (since $x P(1/x) = 1$); the two choices differ only by the on-shell δ -function term. The phrase “set of measure zero” is correct, but it is the prescription, not the measure-zero argument, that makes the integrals well defined.

Comment 3: running coupling and the β -function coefficient in (4.78)

Referee, comment 3

At this order in perturbation theory the QCD coupling should run and the NLO result should depend on a scale associated with the coupling constant renormalization. Indeed one has here a β -function coefficient in (4.78), but somehow no running of the coupling constant. This should be addressed in the text.

Reply. I thank the referee for this remark. First, the result itself does not change. The transverse integral $\int d^2\tilde{p}/\tilde{\mathbf{p}}^2$ in Eq. (4.79) is scaleless, so in dimensional regularisation it vanishes,

$$\mu^{2\epsilon} \int \frac{d^{2-2\epsilon}\tilde{p}}{\tilde{\mathbf{p}}^2} = 0.$$

The purely gluonic overlap Eq. (4.79) therefore does not contribute to the normalisation coefficient, and Eq. (4.80) is correct as it stands.

The referee’s point concerns what this zero means, and I agree it must be stated in the text. The integral vanishes because its UV and IR divergences cancel, not because it is finite:

$$0 = \mu^{2\epsilon} \int \frac{d^{2-2\epsilon}\tilde{p}}{\tilde{\mathbf{p}}^2} = \pi \left(\frac{1}{\epsilon_{\text{UV}}} - \frac{1}{\epsilon_{\text{IR}}} \right) \Big|_{\epsilon_{\text{UV}}=\epsilon_{\text{IR}}}. \quad (6)$$

The prefactor that multiplies this integral in Eq. (4.79) comes from Eq. (4.78) and contains the constant $11/6$. Its UV part is therefore

$$\frac{g^2 N_c}{8\pi^3} \left(-\frac{11}{6} \right) \frac{\pi}{\epsilon_{\text{UV}}} = -\frac{g^2}{16\pi^2} \frac{\beta_0}{\epsilon_{\text{UV}}}, \quad \beta_0 = \frac{11}{3} N_c \quad (N_f = 0).$$

This is the one-loop gluon field-strength renormalisation. In light-cone (axial-type) gauge it also fixes the renormalisation of the coupling ($Z_g = Z_3^{-1/2}$), which is why the β -function coefficient appears in Eq. (4.78). The additional $2 \log(k^+/\Lambda)$ term is the light-cone-gauge rapidity (small- x) divergence. The $1/\epsilon_{\text{IR}}$ pole has collinear origin.

So the vanishing of Eq. (4.79) is the familiar statement that on-shell massless self-energies vanish in dimensional regularisation. The UV divergence has not disappeared: it is absorbed into the renormalised coupling $g(\mu)$ ($\overline{\text{MS}}$). The IR (collinear) divergence cancels against real emission in

infrared-safe quantities. The renormalisation scale μ therefore does not show up in $\mathbb{N}$ at $\mathcal{O}(g^2)$; it enters through the coupling. The $\mathcal{O}(g)$ coefficients $\mathbb{A}$ and $\mathbb{C}$ are tree level and are to be read as functions of $g(\mu)$. Their one-loop, $\mathcal{O}(g^3)$, corrections (Ref. [17]) carry the compensating $\log \mu^2$ with coefficient $\propto \beta_0$. Together these turn the fixed coupling of the Weizsäcker–Williams amplitude into a running coupling evaluated at a scale set by the gluon transverse momentum, as in the running-coupling BK/JIMWLK kernels (Balitsky; Kovchegov and Weigert). The running thus becomes explicit in the physical observable, single inclusive gluon production at NLO [44], which combines the present $\mathcal{O}(g^2)$ coefficients with the $\mathcal{O}(g^3)$ ones. The same remarks apply to Eq. (6.8), which is the one-loop gluon self-energy and likewise vanishes as a scaleless integral.

Proposed changes to the manuscript

1. Keep Eq. (4.80) unchanged. After Eq. (4.79), add a paragraph stating that the scaleless integral vanishes because of the UV/IR cancellation (6). Identify the 11/6 in Eq. (4.78) with $\beta_0/(2N_c)$, and state that the UV part is the coupling (field-strength) renormalisation, absorbed into $g(\mu)$ in $\overline{\text{MS}}$.
2. Below Eq. (5.2), state that $g \equiv g(\mu)$, and that the running coupling appears in physical observables once the $\mathcal{O}(g^3)$ coefficients of Ref. [17] are included. Add a sentence to the conclusions.
3. After Eq. (6.8), add the same remark (scaleless one-loop self-energy).
4. Add references: I. Balitsky, Phys. Rev. D 75 (2007) 014001 [hep-ph/0609105]; Yu. V. Kovchegov and H. Weigert, Nucl. Phys. A 784 (2007) 188 [hep-ph/0609090]; G. Leibbrandt, Rev. Mod. Phys. 59 (1987) 1067 (non-covariant gauges).

Note to the author (not for the referee)

Your treatment is kept: the integral is zero and $\mathbb{N}$ in Eq. (4.80) does not change. Equation (6) is not a new, non-zero contribution. It only splits that zero into its UV and IR parts, to show the referee where the β -function coefficient and the coupling renormalisation sit. If you prefer not to write the $1/\epsilon$ poles at all, a sentence of the form “this zero reflects the cancellation of a UV divergence, which renormalises the coupling (coefficient β_0), against a collinear IR divergence” is enough to answer the referee. Please verify the bibliographic data of the added references.

Comment 4: “operator-valued coefficients” in Sec. 3.1

Referee, comment 4

[...] In Sec 3.1 it is stated that the “coefficients are not ordinary c-number functions but operator quantities acting in the valence Hilbert space”. [...] The full Hilbert space of the theory is of course a complex-coefficient vector space [...]. The statement about noncommuting operator coefficients only applies to the soft gluon part of the Hilbert space, not the full one. The fact that the states in e.g. eq (3.2) are states in only the soft subspace should be emphasized more in the text.

Reply. I agree that the wording was misleading. The full Hilbert space is $\mathcal{H} = \mathcal{H}_{\text{soft}} \otimes \mathcal{H}_{\text{val}}$, an ordinary complex vector space. In any complete basis $|n\rangle_{\text{soft}} \otimes |v\rangle_{\text{val}}$ all expansion coefficients are c-numbers. The coefficients in Eq. (3.2) and Eq. (4.1) are operator-valued only because the expansion is performed in the Fock basis of the *soft* subspace alone, with the valence degrees of freedom left unexpanded; the coefficients then act on $\mathcal{H}_{\text{val}}$ through ρ . The paragraph is rewritten as follows (new text):

Proposed changes to the manuscript

“The expansion (3.2) is an expansion in the Fock basis of the soft subspace $\mathcal{H}_{\text{soft}}$ only. The full Hilbert space $\mathcal{H}_{\text{soft}} \otimes \mathcal{H}_{\text{val}}$ is an ordinary complex vector space, and in a complete basis $|n\rangle \otimes |v\rangle$ all coefficients are c-numbers. We choose not to expand the valence part, which is the natural choice in the Born–Oppenheimer treatment. The coefficients of the soft Fock states are then functionals of the valence charge density ρ , i.e. operators on $\mathcal{H}_{\text{val}}$. Since the ρ ’s obey the algebra (2.19), these operator-valued coefficients do not commute in general, and their ordering must be kept. All states written in the form (3.2), and the vacuum $|0\rangle \equiv |0\rangle_{\text{soft}} \otimes |v\rangle$ (see App. B), should be understood in this sense.”

In addition, the definition $|0\rangle \equiv |0\rangle \otimes |v\rangle$ now given only in App. B is moved into Sec. 3.1. The first sentence of Sec. 4 (“The perturbative coefficients are not ordinary c-number functions...”) is rephrased in the same way.

Part II — Further errors found in the manuscript

The following items were found in a complete re-reading. Group A contains substantive points; the first four affect statements in the paper, not only typography. Group B lists equation-level errors. Group C lists presentation and reference issues. A short version of B and C (“Other changes”) can be appended to the reply.

A. Substantive points

A1. Normalisation of the one-gluon state, Eqs. (4.66)–(4.76).

- (a) The component $|(\Psi_i^a(k))_{gg\rho}\rangle$ appears in Eqs. (4.66), (4.68), (4.70), (4.75), but it is *not defined* in Sec. 4.2. From its use in Eq. (4.75) it must be the disconnected $\mathcal{O}(g)$ emission by the valence, with the incoming gluon as a spectator (the term $\mathbb{A} a^\dagger$, which is omitted from Eq. (4.16)). It should be defined explicitly.
- (b) The statement that $\langle(\Psi_i^{a'}(k'))_{gg\rho}|(\Psi_i^a(k))_{gg}\rangle + \text{h.c.} = 0$ “identically” (Eq. (4.68)) cannot hold as a mutual cancellation. The first term requires $k^+ > k'^+$ (splitting $k \rightarrow k' + (k - k')$ with $(k - k')^+ > 0$), while its partner requires $k'^+ > k^+$. The two terms have disjoint support, so each is separately non-zero. *[Derived from plus-momentum positivity.]*
- (c) Condition Eq. (4.67) with $k \neq k'$ also receives ρ -linear contributions from the overlap of the bare state $|g(k')\rangle$ with the $\mathcal{O}(g^2)$ one-gluon components Eqs. (4.43), (4.44), (4.46), (4.48) and Eq. (4.55). These are not included in Eq. (4.70). The off-diagonal ($k \neq k'$) part of Eq. (4.67) is precisely the unitarity relation Eq. (4.10) for $\mathbb{B}_3$, and should be shown to hold (item (b) terms cancelling against these). Only the $\delta^3(k - k')$ part determines $\mathbb{N}$.

A2. Normalisation of the two- and three-gluon states (Secs. 4.3.3, 4.4.1). The statement “Since the interacting part of the wave function itself starts at order g^2 , corrections to the normalization factor ... contribute only at higher order” is incorrect as written. Eq. (4.17) shows that $\Omega|gg\rangle$ has $\mathcal{O}(g)$ components ($\mathbb{A}^\dagger$ absorption, $\mathbb{A}$ emission, $\mathbb{C}$ splitting, $\mathbb{C}^\dagger$ merging), whose squares contribute at $\mathcal{O}(g^2)$ to the norm. It is only the *new, connected* $\mathcal{O}(g^2)$ components computed in Secs. 4.3–4.4 that do not affect the norm at this order. The same $\mathbb{N}$ multiplies all sectors because Ω is a single operator, and its value is fixed by unitarity. The $\mathcal{O}(g)$ components should also be listed (or referred to) in Eqs. (4.95) and (4.100).

A3. Section 6 (diagonalisation).

- (a) Eq. (6.4): the sign of the first term is wrong. With $\Omega^{(1)} \ni \mathbb{A} a^\dagger - \mathbb{A}^\dagger a$ and $[H_0, a^\dagger(p)] = \frac{\mathbf{p}^2}{2p^+} a^\dagger(p)$ one gets $+\mathbb{A} \frac{\mathbf{p}^2}{2p^+} a^\dagger + \mathbb{A}^\dagger \frac{\mathbf{p}^2}{2p^+} a$. Only with the $+$ sign does the term cancel H_g , using Eq. (5.2): $\mathbb{A} \mathbf{p}^2/2p^+ = -g \mathbf{p}^j \rho(-\mathbf{p})/(\sqrt{2} p^{+3/2})$. The sign convention of the $\mathbb{C}$ terms in the same equation is also consistent only with $+$. [*Derived.*]
- (b) Eq. (6.5) holds only through $\mathcal{O}(g)$. The sentence after it (“the classical WW energy ... is already described in the choice of $\mathbb{A}$ ”) is confusing, because the ρ^2 energy is $\mathcal{O}(g^2)$ and appears only in Eq. (6.10).
- (c) Eq. (6.7) keeps only the vacuum-energy term and one $\mathbb{C}^\dagger \mathbb{C}$ term. At $\mathcal{O}(g^2)$ there are also ρ -linear and $\rho\rho$ terms of the form $a^\dagger a$, from $[H_g, \mathbb{C}]$, $[H_{ggg}, \mathbb{A}]$, $H_{gg\text{-inst}}$ and $[H_g, \mathbb{A}]$, and $a^\dagger a^\dagger a a$ terms, from H_{gggg} , the instantaneous term and $[H_{ggg}, \mathbb{C}]$ including the t -channel. These cancel against $[H_0, g^2 \Omega^{(2)}]$ only *off* the energy shell (Comment 2). The final statement Eq. (6.10) should be qualified accordingly.
- (d) Eq. (6.8) is the one-loop gluon self-energy. It vanishes only as a scaleless integral in dimensional regularisation, and it carries the same UV pole as Eq. (4.79) (Comment 3).

A4. Definition of Ω , Eq. (3.7). As written, $T \exp(-i \int_{-\infty}^0 dt H(t))$ uses the full Schrödinger-picture Hamiltonian and ordinary time t . For the adiabatic theorem/Gell-Mann–Low construction one needs the interaction-picture Hamiltonian with light-cone time and a switching factor, $U_\varepsilon(0, -\infty) = T \exp(-i \int_{-\infty}^0 dx^+ e^{\varepsilon x^+} H_I(x^+))$, and a normalisation that removes the divergent phase. The notation $U(-\infty, 0)$ vs. $U(0, -\infty)$ should also be fixed. This is directly relevant to Comment 2.

A5. Plus momentum carried by ρ . In definition Eq. (2.18) both valence operators carry the same k^+ , so ρ transfers only transverse momentum. Yet Fig. 1 and Eqs. (4.42)–(4.48) assume that the valence exchanges plus momentum with the soft gluon, restricted to $\Lambda < |p^+ - k^+| < \vee$. The paper should explain how the plus-momentum transfer arises from the eikonal Hamiltonian. It should also note that the window $|p^+ - k^+| > \Lambda$ is what regularises the $1/(p^+ - k^+)^2$ singularity of the instantaneous vertex Eq. (D.5), so the results of Eqs. (4.43), (4.44) and (5.4) depend on it. (Fig. 1’s caption mentions only $\Lambda < p^+ - k^+ < \vee$; Eq. (4.44) uses $\Lambda < k^+ - p^+$.)

A6. Power counting. The expansion $\Omega = 1 + g\Omega^{(1)} + g^2\Omega^{(2)}$ counts $\rho = \mathcal{O}(1)$, i.e. the dilute regime. In the saturation regime $\rho \sim 1/g$ and $g\rho = \mathcal{O}(1)$, and the classification into “ $\mathcal{O}(g)$ ” and “ $\mathcal{O}(g^2)$ ” changes (e.g. $g^2\rho\rho$ terms are $\mathcal{O}(1)$). One or two sentences stating the counting would prevent confusion, e.g. with the phrase “at leading order in the coupling and the highest order of ρ ” in Sec. 3.3. The notation should also be made consistent: in Eq. (4.2) the g is factored out, while the footnote states that $\mathbb{A}$ itself is $\mathcal{O}(g)$.

A7. Appendix E. (a) The denominators in Eq. (E.3) contain $E_i^{(1)}$ and $E_i^{(2)}$, which is Brillouin–Wigner-like and not the Rayleigh–Schrödinger theory quoted. (b) The symbol $|n^{(i)}\rangle$ is used both for the i -th order correction and for intermediate basis states. (c) The degenerate case (Comment 2) is not addressed. (d) Despite its title, the appendix does not show where operator ordering matters. For example, $E_i^{(1)}$ is itself an operator on $\mathcal{H}_{\text{val}}$, and Eq. (4.28) is the place where ordering enters. A rewrite along the lines of Eq. (4)/Eq. (5) would serve better.

B. Equation-level errors

Eq.	As printed	Correction	Status
(4.6)	$N_2 = 1 - \int \mathbb{A}^\dagger \mathbb{A}$	$N_2 = -\frac{1}{2} \int \frac{d^3 p}{(2\pi)^3} \mathbb{A}^\dagger \mathbb{A}$ (consistent with the second equality and with Eq. (5.1))	Derived
(4.17), line 1	$-\frac{1}{(2\pi)^{3/2}} \mathbb{A}_j^{\dagger b}(p) g_k^c(q)\rangle$	$\dots g_i^a(k)\rangle$ (absorbing p leaves k)	Derived
(4.17), line 2	$\int \frac{d^3 q}{(2\pi)^3} \mathbb{A}_k^c(q) ggg\rangle$	missing factor $(2\pi)^{3/2}$, since $a^\dagger gg\rangle = (2\pi)^{3/2} ggg\rangle$ by (B.4), (B.5); cf. Eq. (4.15)	Derived
(4.31), (4.36)	$-ig^2 f^{abc} \dots$ (commutator term)	Inserting Eq. (4.29) into Eq. (4.27) gives the same expression with $+ig^2 f^{abc}$ (from $[\rho^b(-\mathbf{k}+\mathbf{p}), \rho^c(-\mathbf{p})] = if^{bcd} \rho^d(-\mathbf{k})$); the sign should be checked	To verify (sign)
(4.36)	sum of (4.31) and (4.22) only	the instantaneous term Eq. (4.24), listed in Eq. (4.35), is missing	Derived
(4.55)	$-\frac{g^2 f^{abc}}{8\pi^3} \int dp^+ \dots$	$+\frac{ig^2 f^{abc}}{8\pi^3} \int \frac{dp^+}{\sqrt{k^+ p^+}} \dots$: the factor $1/\sqrt{k^+ p^+}$ is required by dimensions (compare Eq. (4.54)), and the i comes from $[\rho^b(-\mathbf{p}), \rho^a(\mathbf{k})] = -if^{abc} \rho^c(\mathbf{k}-\mathbf{p})$. Same correction in the second line of Eq. (5.4)	Derived
(4.61)	$\left(\frac{v^++p^+}{q^+} \mathbf{k}^k - \mathbf{p}^k - \mathbf{v}^k\right) \delta_{jn}$	$\mathbf{k}^k \rightarrow \mathbf{q}^k$ (vertex for $v \rightarrow p+q$, cf. Eq. (D.3))	Derived
(4.60)	factor “ $2 \times \frac{1}{12}$ ”	check against (4.62): the matrix element $\langle ppp H_{ggg} uv \rangle$ already contains the splitting of either gluon	To verify
(4.86)	$f^{dbe}, \rho^e(-\mathbf{s})$	f^{dbc} and $\rho^e(\mathbf{s})$ (absorption of s ; cf. Eq. (4.41) and Eq. (4.88))	Derived
(4.94)	$-\frac{g^4 f^{ade} f^{bfh}}{128\pi^3}$; LHS $ (\Psi)_{gg}\rangle$	$g^4 \rightarrow g^2$ (two H_{ggg} vertices); LHS should be $ (\Psi_{ji}^{ba}(k,p))_{gg}^3\rangle$	Derived
(4.97) vs (4.99)	measure $d^3 p$ vs $\frac{d^3 p}{(2\pi)^3}$	use the same measure in both	Derived
(5.3)	$(\mathbf{p}^i + \mathbf{k}^i)$	$(\mathbf{p}^i + \mathbf{q}^i)$; k is not defined here, $k = p + q$ (cf. Eq. (4.22))	Derived
(5.6)	$\frac{\mathbf{q}^2}{2p^+}$ in $\left(\frac{\mathbf{p}^2}{2p^+} + \frac{\mathbf{q}^2}{2p^+} + \frac{\mathbf{r}^2}{2r^+} - \frac{\mathbf{k}^2}{2k^+}\right)$	$\frac{\mathbf{q}^2}{2q^+}$	Derived
(5.7)	$\left(\frac{\mathbf{p}^2}{2p^+} - \frac{\mathbf{q}^2}{2q^+} - \frac{\mathbf{r}^2}{r^+}\right)$	$\frac{\mathbf{r}^2}{2r^+}$	Derived
(5.7)	$\mathbb{C}(r', k, p) \mathbb{C}^\dagger(r', r, q)$	With the convention of Eq. (4.1) (first argument = annihilated gluon), this product describes $r, q \rightarrow r' \rightarrow k, p$, the reverse process. The s -channel term should read $\mathbb{C}(r'; r, q) \mathbb{C}_2(r'; k, p)$ with $\mathbb{C}_2 = -\mathbb{C}^\dagger$ (Eq. (4.8)). Colour/polarisation indices are also missing in (5.7). Add $\mathbb{D}_2^{(t)}$ (Comment 1)	To verify

Eq.	As printed	Correction	Status
(5.7)	sign of the energy factor in the $\mathbb{C}\mathbb{C}^\dagger$ term	it uses $(E_f - E_i)$ in the denominator, while all other terms use $(E_i - E_f)$; check that this matches the sign of $\mathbb{C}_2 = -\mathbb{C}^\dagger$	To verify
(6.4)	$-\mathbb{A}_j^b(p) \frac{\mathbf{p}^2}{2p^+} a_j^{\dagger b}(p)$	$+\mathbb{A}_j^b(p) \frac{\mathbf{p}^2}{2p^+} a_j^{\dagger b}(p)$ (see A3(a))	Derived
(2.13) vs (C.6)	$+\frac{g^2}{2} f f \frac{1}{\partial^+}(\cdot) \frac{1}{\partial^+}(\cdot)$ vs $\frac{g^2}{4} f f(\dots - 2 \frac{1}{\partial^+}(\cdot) \frac{1}{\partial^+}(\cdot))$	opposite signs of the instantaneous four-gluon term; one is a typo (the standard light-cone Hamiltonian has $+\frac{g^2}{2}$, consistent with (C.5))	Derived (inconsistency)
(A.5)	$H_{LCYM} = P^+$	P^- (with Sec. 2.1, p^- generates x^+ translations), or state that $P_+ = P^-$	Derived
(A.1)	$[D_\mu, D_\nu]$	$[D_\mu, D_\nu]$	Typo
(B.5)	$E_{gg}(k, p, q)$	$E_{ggg}(k, p, q)$	Typo
(4.33)	$ \Psi\rangle$	$ \Psi\rangle$	Typo
(4.74), (4.75)	$\mathbf{k}'^{2'}$	$\mathbf{k}'^2$	Typo
(4.53)– (4.55)	superscripts 1, 2 on the $\{\rho, \rho\}$, $[\rho, \rho]$ components in (4.53)	drop them, or use them in (4.54), (4.55) as well	Typo
(2.21), (2.23)	stray commas “+ h.c.”; unbalanced bracket in (2.23)	remove / balance	Typo

C. Presentation and references

- C1.** Sec. 4.3: the incoming momenta are labelled (k, p) in (4.81), (p, q) in (4.83) and (r, s) in (4.85)–(4.92), but summed as (k, p) in Eq. (4.95). Use one labelling throughout.
- C2.** The instantaneous four-gluon Hamiltonian is called $H_{gggg\text{-inst}}$ in (2.17), (2.24), (6.6) and $H_{gg\text{-}gg\text{ inst}}$ in (4.81), (D.7). Use one name.
- C3.** Sec. 4.2.5, before Eq. (4.74): the sentence “The only remaining contribution requiring special attention is the overlap of the purely two-gluon sector” is placed before the ρ -overlaps. It belongs before Eq. (4.77).
- C4.** Sec. 5, first paragraph: the matching also uses $\Omega|ggg\rangle$ (for $\mathbb{E}_2$), not only $\Omega|0\rangle, \Omega|g\rangle, \Omega|gg\rangle$.
- C5.** Introduction, last paragraph: “diagonalize the QCD Hamiltonian” should be “pure Yang–Mills”; Sec. 7 is not described.
- C6.** Eq. (5.11): the phrase “resumming repeated gluon emissions . . . into a single unitary operator” overstates what a truncated $\mathbb{G}$ does. Suggest “ $\mathbb{G}$ is the Hermitian generator of Ω , known here through $\mathcal{O}(g^2)$ ”.

C7. References:

- 20 and [32] are the same paper (Altinoluk & Kovner, PRD 83 (2011)); [38] and [41] are the same paper (Munier, arXiv:2510.05256). Merge them and renumber.
- Introduction: “[20–24, 24–33]” cites [24] twice; write “[20–33]”.
 - 7, [8]: “the The Color Glass Condensate” $\rightarrow$ “the Color Glass Condensate”.
 - 53 is Peskin & Schroeder (co-author missing). For the treatment of scaleless integrals a more specific reference (e.g. J. C. Collins, *Renormalization*, CUP 1984) is preferable.
 - 50 : “Lifshits” $\rightarrow$ “Lifshitz”.

- Several journal references lack article/page numbers ([10]–[12], [15], [16], [18]–[23], [25], [27], [29], [31], [35], [36]).
- Conclusions: “...requires the explicit coefficients of the evolution operator derived here [54]”. [54] is Kovner & Lublinsky (2006); the intended reference is probably [44]. Please check.
- Add the running-coupling and non-covariant-gauge references listed under Comment 3.

With these changes I believe all the referee’s concerns have been addressed. I thank the referee again for comments that have improved the manuscript.

Sincerely,

Ramkumar Radhakrishnan